

INSTRUCTFX2FX: A MULTI-TURN TEXT-TO-PRESET DEMO FOR ITERATIVE AUDIO EFFECT REFINEMENT

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ABSTRACT

We present *InstructFX2FX*, an interactive demonstration of multi-turn, text-guided audio effect refinement. Existing text-to-preset systems are largely single-shot: given one textual descriptor, they generate one preset. Real audio engineering, however, is sequential: engineers iteratively refine an existing effect chain through successive instructions. This raises a stateful problem: given the current effect parameters and a sequence of instructions, how can a system update the sound while preserving what the previous instructions already achieved?

InstructFX2FX addresses this problem with a hybrid language-model and CLAP-optimization architecture. An LLM acts as a high-level planner, selecting effects, ordering the signal chain, and proposing initial parameters — motivated by recent evidence that LLM-based initialization outperforms CLAP optimization from random initialization [1]. CLAP-guided optimization then refines those parameters perceptually, providing a more reproducible refinement signal than re-prompting the LLM at each turn.

At the demo, attendees drive a dry recording through successive natural-language instructions and inspect the evolving FX chain, intermediate optimization checkpoints, and session state. On SocialFX-derived descriptor transitions, CLAP-guided refinement consistently lowers target-directed DSP-feature MMD relative to an LLM-only reprompting baseline, reducing the average across directed descriptor pairs by roughly 24% (0.45 to 0.34).

1. INTRODUCTION

Audio effect design is rarely a one-shot process. In practice, audio engineers and producers establish a rough effect chain and then iteratively refine it through feedback such as “make it warmer,” “add some weight,” or “too harsh, soften it.” Each instruction is interpreted relative to the current sound and parameter state, not as an independent request from scratch.

We call this problem *sequential FX refinement*. Given an existing parameter state P and a sequence of natural-language instructions $\{I_1, I_2, \dots\}$, the goal is to update the effect parameters so that the rendered audio tracks the user’s evolving intent while preserving relevant structure from the existing state. This differs from conventional text-to-preset generation, where each prompt is independent.

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Recent text-guided systems address only the single-turn case. Text2FX-style methods optimize differentiable parameters using CLAP-guided objectives from a random initialization [2]. LLM-based systems such as LLM2Fx predict parameters directly from text and have shown that language-model initialization can outperform CLAP refinement from random starts [1]. FxSearcher extends CLAP-guided search to non-differentiable plugins via Bayesian optimization [3]. None of these directly support sequential refinement, and naïvely re-prompting an LLM at each turn is unreliable: the model regenerates parameters non-deterministically, perturbs settings the user did not ask to change, and — most fundamentally — cannot hear the rendered audio.

We therefore propose a hybrid design. The language model acts as the “brain” of the system: it selects which effects to apply, orders the signal chain, and generates reasonable initial parameters. CLAP-guided optimization then serves as the perceptual refinement mechanism by iteratively adjusting the effect parameters so that the rendered audio embedding better aligns with the user instruction embedding in CLAP space. In this view, the LLM provides the starting point, while CLAP provides the refinement direction.

We implement this in *InstructFX2FX*, a session-aware demonstration system for multi-turn text-guided audio effect refinement. The system maintains a persistent session state containing the current effect chain, parameter values, rendered audio, and instruction history. At each turn, it decides whether to reuse existing effects, initialize new effects, or jointly refine a mixed chain. For differentiable effects such as EQ and reverb, parameters are refined with CLAP-guided gradient descent. For non-differentiable effects such as compressor, distortion, delay, bitcrush, and pitch shifting, the system uses Bayesian optimization with a CLAP-based objective.

The contribution of this demo paper is therefore a formulation and system demonstration for stateful, multi-turn audio effect refinement.

2. PROBLEM FORMULATION

We formulate sequential FX refinement as a stateful parameter-update problem. Let x denote the dry input audio, C_t the current effect chain at turn t , and P_t the corresponding effect parameters. Given a user instruction I_t , the system produces an updated chain and parameter state:

$$(C_t, P_t) = f(x, C_{t-1}, P_{t-1}, I_t, H_{t-1}),$$

where $H_{t-1} = \{I_1, \dots, I_{t-1}\}$ denotes the prior instruction history.

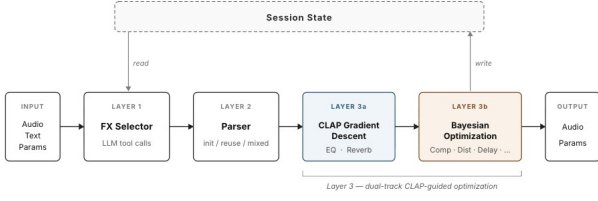


Figure 1: The InstructFX2FX harness. The LLM planner selects and orders the FX chain, the routing module determines how the current session state should be updated, and the optimization backend refines parameters in CLAP space. The persistent session state makes the system iterative.

This formulation differs from single-shot text-to-preset generation:

$$P = g(x, I),$$

where each prompt is processed independently. In sequential refinement, the desired behavior is not merely to generate a plausible preset for the latest instruction, but to update the existing sound in a controlled direction. For example, after the sequence “make it warm” → “make it heavier,” the second instruction should preserve the warmth already introduced while moving the sound toward heaviness.

This distinction creates three requirements. First, the system must maintain persistent state across turns. Second, it must decide whether a new instruction should reuse existing effects, introduce new effects, or both. Third, it must refine parameters according to the actual rendered audio, because the same parameter change may have different perceptual consequences depending on the current chain and input signal.

3. SYSTEM OVERVIEW

Figure 1 shows the InstructFX2FX architecture. The system takes a dry audio signal, a user instruction, and the current session state as input. The session state stores the active effect chain, parameter values, rendered checkpoints, and prior instruction history.

The architecture separates high-level semantic planning from perceptual parameter refinement. The LLM planner is responsible for symbolic decisions such as selecting effects, ordering the signal chain, and proposing initial parameter values. CLAP-guided optimization then refines the selected parameters using the rendered audio as perceptual feedback.

This separation is especially important in the multi-turn setting. Instead of regenerating presets independently at every turn, the system performs local updates relative to the current rendered sound and parameter state.

3.1. Hybrid LLM + CLAP Design

The system combines two complementary components. The language model provides a strong semantic prior by mapping textual instructions to plausible effect choices and parameter ranges. This allows the system to initialize or extend an effect chain without searching the full parameter space from scratch.

However, language models alone are poorly suited to sequential refinement. Re-prompting may regenerate unrelated presets, perturb parameters that should remain stable, or fail to account

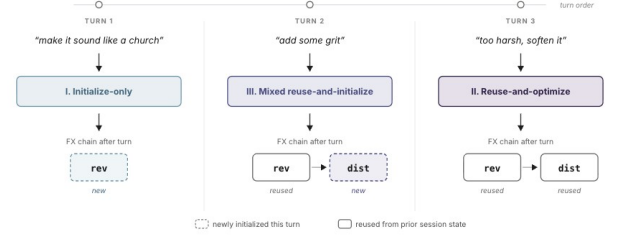


Figure 2: The three routing modes used in multi-turn refinement. Initialize-only creates a new chain, mixed reuse-and-initialize extends the existing chain, and reuse-and-optimize refines the current chain in place.

for the actual rendered audio. Since the LLM does not directly perceive the current sound, it cannot reliably determine whether a refinement step improved the audio perceptually.

CLAP-guided optimization provides this missing feedback signal. Instead of predicting parameters from text alone, the system renders candidate audio and evaluates its semantic alignment with the target instruction in audio-text embedding space. The optimization stage therefore acts as a perceptual refinement layer on top of the LLM-generated initialization and the current session state.

3.2. Session-aware Multi-turn Refinement

A central requirement of sequential FX refinement is that each instruction should operate on the existing state. InstructFX2FX therefore maintains a persistent session state across prompts. At every turn, the system reads the current effect chain, parameter values, and instruction history before deciding how to respond.

The routing module dispatches each instruction into one of three execution modes:

- **Initialize-only:** no existing effects are present, so the LLM initializes a new chain and parameters.
- **Reuse-and-optimize:** all relevant effects already exist, so the system refines the current parameters in place.
- **Mixed reuse-and-initialize:** part of the existing chain is reused, while newly required effects are initialized and inserted.

Using Figure 2 as an example, the instruction sequence “make it sound like a church” → “add some grit” → “too harsh, soften it” triggers three different behaviors. The first turn initializes a reverb effect, the second adds distortion on top of the existing reverb chain, and the third performs corrective refinement on the current chain without introducing additional effects.

Thus, corrective instructions are treated as local updates to the current sound, while additive instructions can extend the chain when needed.

3.3. Optimization Backends

After the LLM planner determines the effect chain and initial parameters, InstructFX2FX refines the selected parameters using existing CLAP-guided optimization objectives. Our contribution is not a new optimizer, but a session-aware framework that integrates

Table 1: Effects currently supported in the demonstration system.

Category	Effects	Optimization
Diff.	EQ, reverb	Gradient descent
Non-diff.	Compressor, distortion, delay, bitcrush, pitch shift	Bayesian optimization

these optimization methods into an interactive multi-turn workflow. Table 1 summarizes the effects currently supported in the demonstration system and their corresponding optimization backends.

For differentiable effects, we adopt existing CLAP-guided objectives used in prior text-guided FX optimization work. Let $\phi_{\text{audio}}(\cdot)$ and $\phi_{\text{text}}(\cdot)$ denote the CLAP audio and text embedding functions.

The semantic similarity objective directly aligns rendered audio with the target instruction:

$$\mathcal{L}_{\text{sem}} = 1 - \cos(\phi_{\text{audio}}(\mathbf{x}), \phi_{\text{text}}(I)). \quad (1)$$

For sequential refinement, we additionally use a directional objective that encourages the refinement trajectory in audio-embedding space to follow the semantic transition implied by the instruction:

$$\mathcal{L}_{\text{dir}} = 1 - \cos(\phi_{\text{audio}}(\mathbf{x}(t)) - \phi_{\text{audio}}(\mathbf{x}_0), \phi_{\text{text}}(B) - \phi_{\text{text}}(A)). \quad (2)$$

We also support guided semantic objectives with negative anchors following prior CLAP-guided optimization work:

$$\mathcal{L}_{\text{guided}} = 1 - \cos(\phi_{\text{audio}}(\mathbf{x}), \phi_{\text{text}}(I^+)) + \cos(\phi_{\text{audio}}(\mathbf{x}), \phi_{\text{text}}(I^-)). \quad (3)$$

In both cases, optimization starts from the LLM-generated or session-reused parameter state. The optimization stage therefore acts as a perceptual refinement layer on top of the evolving session state.

4. EVALUATION

We conduct a preliminary evaluation to test whether InstructFX2FX improves over LLM-only sequential reprompting after a reasonable language-model initialization. The evaluation focuses on EQ-related descriptor transitions derived from the SocialFX dataset [4].

4.1. Dataset and Ground Truth

We evaluate on the SocialFX dataset, selecting seven EQ descriptor words by frequency (≥ 20 occurrences) and consistency (≥ 0.45): *warm* (93), *bright* (74), *soft* (74), *loud* (54), *harsh* (39), *calm* (26), and *heavy* (25). From these we construct 10 directed word pairs (e.g., *warm* $\rightarrow$ *bright*, *harsh* $\rightarrow$ *soft*).

Ground truth audio is produced sequentially: for a pair $A \rightarrow B$, we apply the SocialFX EQ parameters for word A to dry audio, then apply the parameters for word B . For each word and instrument condition, we sample 50 random parameter instances.

4.2. Metric: Maximum Mean Discrepancy

All comparisons use Maximum Mean Discrepancy (MMD) with a Gaussian kernel (median bandwidth heuristic) over 35-dimensional DSP feature vectors (spectral centroid, bandwidth, rolloff, MFCCs,

temporal features). Lower MMD indicates that the system output distribution is closer to the sequential ground truth distribution.

This metric is intentionally separate from the CLAP objective used during optimization. Therefore, improvement in MMD indicates that CLAP-guided refinement transfers to a DSP-feature-based evaluation.

4.3. Experiment 1: Sequential MMD

We compare InstructFX2FX against a LLM-only baseline (LLM reprompting without CLAP optimization) on all 10 directed word pairs. As Figure 3a, our system achieves lower MMD on **9 of 10** pairs, corresponding to **90%** of pairs improved. CLAP gradient descent consistently outperforms LLM-only reprompting for EQ-centric descriptors. The largest gains occur on pairs with clear timbral contrast (e.g., *heavy* $\rightarrow$ *calm*, *harsh* $\rightarrow$ *soft*); the smallest gains are on pairs where the CLAP objective and DSP-feature MMD are less aligned (e.g., *calm* $\rightarrow$ *loud*).

4.4. Experiment 2: MMD Convergence Trajectory

We plot MMD against optimization iteration for representative pairs. For *loud* $\rightarrow$ *heavy*, both instruments stay below the LLM-only baseline throughout optimization. For *calm* $\rightarrow$ *loud*, violin initially exceeds the baseline for approximately 0–400 iterations before recovering. This reveals a recurring failure mode: early CLAP progress overshoots the DSP-feature optimum, suggesting a mismatch between the CLAP objective and the MMD evaluation metric.

4.5. Experiment 3: LLM Initialization Ablation

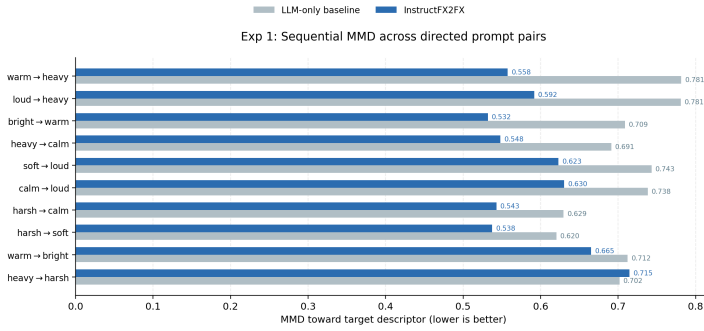
Starting from the same LLM-initialized parameters, we compare final MMD with and without CLAP gradient descent. CLAP refinement improves **87.5%** of word pairs, with an average improvement of approximately 13 MMD units by convergence. The LLM-only condition is flat across iterations, confirming that CLAP GD provides reliable refinement signal beyond the initialization.

5. LIMITATIONS AND FUTURE WORK

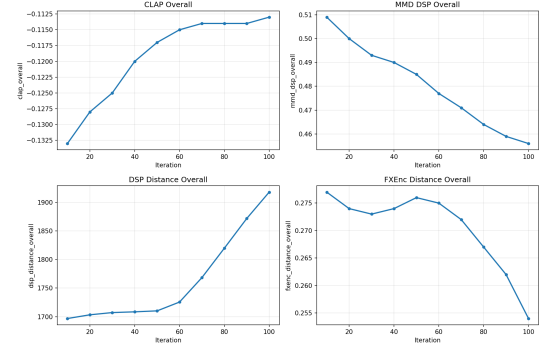
Current limitations include several optimization and evaluation challenges. First, the current quantitative evaluation is centered primarily on EQ-related descriptors because SocialFX provides cleaner descriptor-to-parameter associations for EQ than for more complex effects.

Second, CLAP embedding space does not fully capture all perceptual properties relevant to audio production. Although CLAP-guided refinement often improves early optimization behavior, optimization trajectories may later drift or overshoot, and improvements in CLAP space do not always correspond to improvements in DSP-feature metrics or perceptual quality.

Third, non-differentiable effects remain difficult to refine reliably. Bayesian optimization enables support for Pedalboard-based effects such as distortion, delay, and bitcrushing, but the resulting optimization trajectories are often noisy and semantically inconsistent across iterations. Corrective prompts such as “soften it” after distortion are particularly unstable, suggesting that CLAP-based black-box objectives alone may be insufficient for robust sequential refinement.

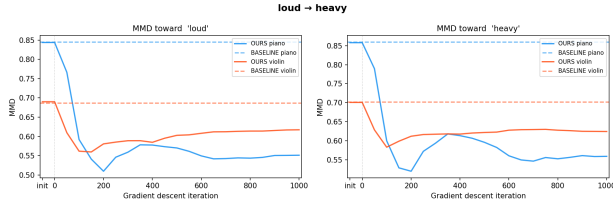


(a) Exp 1: sequential MMD across prompt pairs.

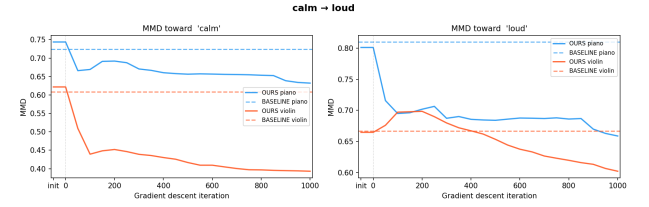


(b) Exp 3: average

Figure 3: Two summary results from the presentation narrative. Left: in Exp 1, InstructFX2FX improves target-directed MMD on most directed prompt pairs relative to an LLM-only baseline. Right: in Exp 3, CLAP refinement improves over the same LLM initialization rather than merely benefiting from a better starting point. In the repository artifacts, overall DSP-feature MMD decreases from 0.4461 to 0.3380, and a March 25, 2026 iteration-analysis artifact reports a 7/8 improvement rate with an average of 12.86 iterations before improvement.



(a) Good case: *loud*→*heavy*.



(b) Messy case: *calm*→*loud*.

Figure 4: Exp 2: representative optimization trajectories. Some prompt pairs show an early drop in MMD and stay below the baseline, while others overshoot or drift later in optimization. These plots motivate why the demo interface exposes intermediate checkpoints instead of only the final render.

Finally, optimization currently requires several seconds per interaction turn, preventing real-time deployment inside production DAW workflows.

Future directions include extending evaluation to richer multi-effect chains, conducting human listening studies for perceptual validation, developing domain-adapted audio-effect embedding models, improving stability for non-differentiable refinement, and integrating the system as a real-time VST/AU plugin.

6. CONCLUSION

We presented InstructFX2FX, a session-aware demonstration system for sequential text-guided audio effect refinement. Unlike single-shot text-to-preset systems, InstructFX2FX treats audio effect editing as a stateful process in which each instruction updates an existing chain and parameter state.

The core idea is a hybrid division of labor: the LLM provides high-level effect planning and initialization, while CLAP-guided optimization provides perceptual refinement. Preliminary experiments on SocialFX-derived descriptor transitions suggest that CLAP-based refinement can improve over LLM-only reprompting in many sequential settings, while also exposing important limitations such as CLAP/DSP-feature mismatch, optimization drift, and instability for non-differentiable effects.

More broadly, this demo is intended as a proof-of-concept for a research direction combining different sources of artificial intelligence toward interactive audio-effect systems that better reflect practical creative workflows.

In particular, our results suggest several promising directions, including audio-effect-specific embedding models, optimization methods designed explicitly for sequential refinement, and generative approaches such as flow-matching-based parameter refinement models trained directly on iterative FX editing trajectories.

7. REFERENCES

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